5 Solve the simultaneous equations

$$4x + y = 10$$
$$x - 5y = 13$$

First equation ×5:

$$\begin{array}{rrcl} & 20x + 5y & = & 50 \\ + & x - 5y & = & 13 \\ \hline & 21x & = & 63 \\ & \div 21 & & \div 21 \\ & x & = & 3 \end{array}$$

1 mark
Eliminate y by addition

1 mark
x = 3; y = −2

Substitute $x = 3$: $y = 10 - 4 \times 3 = -2$.

1 mark
Substitute the found x

Check: $4 \times 3 - 2 = 10$; $3 - 5(-2) = 13$.

$x =$ .....

$y =$ .....

(Total for Question 5 is 3 marks)